HSBI — Bielefeld University of Applied Sciences and Arts

Quantitative Research Methods

Summer Semester 2026 · Examiner: Prof. Dr. Christoph Weisser

Mock Examination 1 (Lectures 1–4)

ISLP Chapters 1–3: Statistical Learning · K-Nearest Neighbours · Linear Regression

Duration:	**90 minutes**
Total credit:	**90 points**
Allowed aids:	non-programmable calculator; one handwritten A4 sheet of notes (both sides)

Name: ______________________________ Matriculation no.: ____________________

Problem	**P1**	**P2**	**P3**	**P4**	**P5**	**Total**
Maximum points	15	15	12	24	24	**90**
Achieved points						

Instructions

- Justify all answers. Numerical results without a derivation or a brief reasoning receive no credit.
- Round final numerical results to 3 decimal places unless stated otherwise. Small deviations caused by carrying rounded intermediate results are accepted.
- The number of points per problem roughly equals the recommended working time in minutes.

Problem 1: Fundamentals of statistical learning (15 points)

(a) **[4 P]** Explain the difference between using a statistical learning method for *prediction* and for *inference*. Give one concrete example of a question for each of the two goals.

(b) **[4 P]** Contrast *parametric* and *non-parametric* methods for estimating the unknown function f in $Y = f(X) + \varepsilon$: describe the two approaches briefly and state one advantage of each. Which general trade-off between *flexibility* and *interpretability* lies behind this choice?

(c) **[3 P]** Classify each of the following scenarios as *regression*, *classification* or *unsupervised learning*, with a one-sentence justification:

(i) A telecom provider has monthly usage, contract length and age for 800 customers and wants to predict whether each customer will cancel the contract next month.
(ii) An economist records profit, number of employees and industry for 90 firms in order to understand which of these factors are associated with CEO salary, and how strongly.
(iii) An online retailer has the shopping baskets of 5,000 customers and wants to discover groups of customers with similar purchasing behaviour.

(d) **[4 P]** Explain why the training MSE of a method is in general *smaller* than its test MSE. What is *overfitting*, and how does it show up when training and test MSE are plotted against model flexibility?

Problem 2: Bias–variance decomposition (15 points)

For the prediction of a response at a fixed test point x_0, three methods of increasing flexibility were analysed. A simulation study yielded the following values (all in squared units of Y); the noise variance is $\mathrm{Var}(\varepsilon) = \sigma^2 = 1.00$:

	Method A (linear)	**Method B** (moderately flexible)	**Method C** (very flexible)
$[\mathrm{Bias}(\hat{f}(x_0))]^2$	2.25	0.16	0.01
$\mathrm{Var}(\hat{f}(x_0))$	0.25	1.44	3.60

(a) **[4 P]** State the decomposition of the expected test MSE $E\Big[(y_0 - \hat{f}(x_0))^2\Big]$ into its three components and explain in one sentence each what the three components measure.

(b) **[5 P]** Compute the expected test MSE at x_0 for each of the three methods. Which method should be chosen, and why?

(c) **[3 P]** What is the smallest expected test MSE at x_0 that *any* method (however good) could achieve here? Justify your answer.

(d) **[3 P]** Describe how squared bias, variance, training MSE and expected test MSE typically behave as the flexibility of a method increases, and explain how the numbers in the table illustrate the characteristic U-shape of the test MSE.

Problem 3: K-nearest neighbours classification (12 points)

An online shop wants to predict whether a visitor makes a purchase. For seven past visitors, two standardised activity scores X_1 (pages viewed) and X_2 (minutes on site) and the outcome Y (purchase: Yes/No) were recorded:

Obs.	X_1	X_2	Y
1	4	3	Yes
2	2	2	No
3	6	2	No
4	4	5	No
5	1	1	Yes
6	7	4	Yes
7	5	6	No

A new visitor has scores $x_0 = (X_1, X_2) = (4, 2)$.

(a) **[4 P]** Compute the Euclidean distance between x_0 and each of the seven observations (3 decimals; you may rank by squared distances). How is the new visitor classified with $K = 1$?

(b) **[4 P]** How is the new visitor classified with $K = 3$? State also the estimated conditional probability $\widehat{\Pr}(Y = \text{Yes} \mid X = x_0)$ that the $K = 3$ classifier uses.

(c) **[2 P]** How does the choice of K affect the bias and the variance of the KNN classifier (small K vs. large K)? What is the *training* error rate of the 1-nearest-neighbour classifier?

(d) **[2 P]** What is the *Bayes classifier*, and why can it not be used directly in practice? In what sense is KNN related to it?

Problem 4: Simple linear regression by hand (24 points)

A café chain investigates the effect of local radio advertising on sales. For $n = 5$ weeks it records the number of radio spots per week x and the weekly sales y (in 1,000 EUR):

Week i	1	2	3	4	5
Spots x_i	1	2	3	4	5
Sales y_i	6	7	11	13	18

You may use without proof: $\sum x_i = 15$, $\sum y_i = 55$, $\sum x_i^2 = 55$, $\sum x_i y_i = 195$. Consider the model $Y = \beta_0 + \beta_1 X + \varepsilon$.

(a) **[6 P]** Compute the least-squares estimates $\hat{\beta}_1$ and $\hat{\beta}_0$, and write down the fitted regression line. Interpret both estimates in the context of the problem (with units).

(b) **[4 P]** Compute the fitted values $\hat{y}_i$ and the residuals e_i for all five weeks, and verify that $\sum_{i=1}^{5} e_i = 0$.

(c) **[3 P]** Compute the residual sum of squares (RSS) and the residual standard error (RSE), and interpret the RSE.

(d) **[6 P]** Compute $\mathrm{SE}(\hat{\beta}_1) = \mathrm{RSE} / \sqrt{\sum_i (x_i - \bar{x})^2}$ and test $H_0\colon \beta_1 = 0$ against $H_1\colon \beta_1 \neq 0$ at level $\alpha = 0.05$ using the t-statistic. Also give the 95 % confidence interval for β_1. *Use* $t_{0.975,3} = 3.182$ *(97.5 % quantile of the t-distribution with 3 degrees of freedom).* State your conclusion in one sentence.

(e) **[3 P]** Compute the total sum of squares (TSS) and the coefficient of determination R^2, and interpret R^2.

(f) **[2 P]** Predict the weekly sales for a week with $x = 6$ radio spots. Name one reason why this particular prediction should be treated with caution.

Problem 5: Multiple linear regression — interpreting Python output (24 points)

A consultancy analyses the salaries of $n = 60$ employees of a software company. The variables are `salary` (annual salary in 1,000 EUR), `experience` (professional experience in years) and the dummy variable `female` (= 1 for female employees, = 0 for male employees). The model

$$\texttt{salary} = \beta_0 + \beta_1\,\texttt{experience} + \beta_2\,\texttt{female} + \beta_3\,(\texttt{experience} \times \texttt{female}) + \varepsilon$$

was fitted by least squares in Python (`statsmodels`), with the following output:

	coef	std err	t	$P > \|t\|$
Intercept	46.2000	1.100	42.000	<0.0001
experience	2.3500	0.250	9.400	<0.0001
female	−3.1000	1.550	−2.000	0.0504
experience:female	−0.8500	0.350	−2.429	0.0184

$$n = 60; \quad R^2 = 0.712; \quad \text{adjusted } R^2 = 0.697;$$
$$F\text{-statistic} = 46.15 \text{ on 3 and 56 degrees of freedom, } \text{Prob}(F\text{-statistic}) < 0.0001.$$

(a) **[6 P]** Interpret all four estimated coefficients precisely in the context of the problem (with units), paying particular attention to the dummy variable and the interaction term.

(b) **[4 P]** Which coefficients are significantly different from zero at level $\alpha = 0.05$? Verify for the interaction term that the reported t-value is consistent with the reported coefficient and standard error. Should `female` be removed from the model because of its p-value? Discuss briefly.

(c) **[4 P]** Write down the fitted regression equation separately for men and for women.

(d) **[4 P]** Predict the annual salary of (i) a man with 10 years of experience and (ii) a woman with 10 years of experience. What is the estimated salary gap between women and men at 10 years of experience, and how can it be written in terms of $\hat{\beta}_2$ and $\hat{\beta}_3$?

(e) **[3 P]** Interpret R^2. State the null hypothesis of the reported F-test and the conclusion that follows from the output.

(f) **[3 P]** Model diagnostics:

- (i) A plot of the residuals against the fitted values shows a clear U-shape (positive residuals for small and large fitted values, negative in between). What does this indicate, and name one possible remedy.
- (ii) In an extended model, the predictor `age` has a variance inflation factor of VIF = 12. What does this mean, and what is the practical consequence for the regression results?

End of the examination — good luck!